

IS-001 Sentinel Kernel

Implementation Specification IS-001 - Sentinel Kernel

Status: Approved for Design

Priority: Critical

Package ID: PKG-001

Python package: sentinel-kernel

Depends on: None

Referenced Blueprint Volumes: 34, 56, related 3, 33, 55

Purpose

The Sentinel Kernel is not a security engine. It is the runtime that coordinates every security engine.

Responsibilities

- Engine lifecycle
- Service registration
- Object registration
- Event routing
- Scheduler
- Permissions
- Configuration
- Health monitoring
- Plugin management
- Dependency management

Non-Responsibilities

The kernel must never scan endpoints, analyze malware, calculate trust, inspect cloud resources or parse network traffic.

Package Structure

sentinel-kernel/

- kernel.py

- runtime.py
- lifecycle.py
- registry.py
- scheduler.py
- permissions.py
- configuration.py
- health.py
- plugins.py
- services.py
- dependency.py
- exceptions.py
- types.py
- interfaces.py
- constants.py

Runtime Lifecycle

Created -> Initialize -> Load Configuration -> Load Plugins -> Register Services -> Start Engines -> Verify Health -> Running -> Maintenance -> Shutdown -> Terminated

Engine Contract

Every engine implements initialize(), start(), health(), pause(), resume(), stop(), shutdown().

Requirements

REQ-KERNEL-001: The kernel shall register services dynamically.

REQ-KERNEL-002: The kernel shall manage engine lifecycles.

REQ-KERNEL-003: The kernel shall validate dependencies before startup.

REQ-KERNEL-004: The kernel shall expose health information.

REQ-KERNEL-005: The kernel shall support plugin registration.

REQ-KERNEL-006: The kernel shall maintain structured logs.

REQ-KERNEL-007: The kernel shall remain independent of security engines.